**Algorithm 1:** ZKP-Based Mineral Extraction Process

**Require:** Input batch data $B_i = \{ID_i, M_i, T_i, E_i\}$;
Private origin coordinates $L_i = (x_i, y_i)$;
Certification attributes $\mathcal{A}$;
Certified region boundary $R$;
Blockchain ledger $\mathcal{B}$;
Smart contracts $SCs$

**Ensure :** Only mineral batches with valid region-compliance and certification proofs are registered on the blockchain as verified extraction records.

`// Step 1: Data Input and Initialization`

Receive mineral batch data $B_i = \{ID_i, M_i, T_i, E_i\}$ for processing

Extract private origin coordinates $L_i = (x_i, y_i)$ from the secure sensor module

Identify specific certification attributes (e.g., labor standards, environmental impact) associated with $B_i$

`// Step 2: Commitment to Sensitive Attributes (Privacy Layer)`

Initialize random blinding factors $r_i$ and $r_i'$ to ensure commitment hiding properties

Generate Pedersen Commitment for origin: $C_{orig} = Com(L_i, r_i)$

Generate Pedersen Commitment for certification: $C_{cert} = Com(E_i, r_i')$

`// Step 3: Proof Generation (Cryptographic Layer)`
`// Utilizing IEEE 754 Floating-Point Arithmetic for GPS precision`

Generate region-compliance proof $\pi_R$ such that the prover knows $L_i$ where $L_i \in R$

Generate certification-validity proof $\pi_E$ such that the prover knows $E_i$ where $E_i = 1$

Construct the complete proof package $P_i = \{C_{orig}, C_{cert}, \pi_R, \pi_E\}$ for on-chain submission

`// Step 4: Smart Contract Submission and Verification`

Transmit metadata $\{ID_i, M_i, T_i\}$ and proof package $P_i$ to the $SC_{extraction}$ validator

Execute on-chain verification for region-compliance: $Verify(C_{orig}, \pi_R, R)$

Execute on-chain verification for certification-validity: $Verify(C_{cert}, \pi_E)$

`// Step 5: Decision and Blockchain Registration`

**if** $Verify(C_{orig}, \pi_R, R) = 1 \wedge Verify(C_{cert}, \pi_E) = 1$ **then**

- Register batch as a verified extraction record on the immutable ledger: $\mathcal{B} \leftarrow SCs(B_i, \tau_{extract})$
- Mint a non-fungible token (NFT) $Token_{batch}$ representing unique ownership and provenance
- Compute $Hash(B_i)$ and store $(Hash(B_i), RegionID, Hash(\pi))$ on-chain
- Off-load the full proof $\pi$ to IPFS for distributed availability
- Emit the $EventExtracted(ID_i, RegionID, Hash(C_{orig}))$ signal to the network
- Set the global batch status to **Verified**

**else**

- Reject the mineral batch and trigger an automated security alert
- Set the global batch status to **Rejected / Flagged for Audit**